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Article

Field Methods
2014, Vol. 26(3) 284-301
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DOI: 10.1177/1525822X13519561
fm.sagepub.com



The State of Survey Methodology: Challenges, Dilemmas, and New Frontiers in the Era of the Tailored Design

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Abstract

In this overview, we discuss the current state of survey methodology in a form that is useful and informative to a general social science audience. The article covers existing challenges, dilemmas, and opportunities for survey researchers and social scientists. We draw on the most current research to articulate our points; however, we also speculate on both technological and cultural changes that currently influence or may soon affect the efficacy of different methodological choices.

Keywords

web surveys, mixed-mode surveys, self-administered surveys, online panels, random digit dial

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Introduction

The remarkable contribution that probability-based sample surveys make to our knowledge stems from their statistical power. By asking questions of only a few hundred or thousand members of a carefully defined population, we can make fairly precise estimates of the distribution of behaviors and opinions of thousands or millions of people (or units) in that population. This ability to generalize with statistical confidence based on probability theory tends to separate surveys from other commonly used research methods such as focus groups, ethnographic methods, or cognitive interviews for collecting information about people's personal characteristics, attitudes, and behaviors (Kish 1965; Valliant et al. 2000).

Still, these estimates are subject to sampling error (the precision with which sampled individuals represent the target population) as well as the other sources of error that surveyors must take into account. They include coverage error (some members of the population do not have a known, nonzero chance of being included in the sample); nonresponse error (respondents to the survey request are different from those who do not respond to that request in ways relevant to the study objectives); measurement error (respondents provide inaccurate answers, commonly the result of poor question wording, mode effects, and respondent attributes); and processing errors (coding and inaccurate data entry).

Although probability sample surveys have stood the test of time, the use of particular modes has changed. For most of the 20th century, the in-person interview was the dominant mode of data collection and for a time was considered the only acceptable method for most studies of households and individuals. For reasons of cost and speed, telephone interviews replaced most in-person data collection in the last quarter of the century. However, over the past decade, a near perfect storm of societal changes has developed, with the result that no generally accepted individual mode of surveying now dominates.

In this overview, our goal is to discuss the current state of survey methodology in a form that is useful and informative to a general social science audience. The article begins where Dillman's 2002 American Association for Public Opinion Research Presidential Address ended, by discussing today's challenges, dilemmas, and opportunities for survey researchers and social scientists (Dillman 2002). When possible, we draw on the most current research to articulate our points; however, just as Dillman did in 2002, we speculate on both technological and cultural changes that currently influence or may soon affect the efficacy of different methodological choices.

The Limitations of Individual Modes

Interviews

Researchers have long recognized that respondents to interviews, whether in-person or by telephone, give answers that are influenced by the interviewer. They are more likely to give socially desirable answers (e.g., "Yes, I voted in the last election") and less likely to give socially undesirable answers (e.g., "No, I do not regularly drive my car after having several alcoholic drinks;" Belli et al. 1999; Davis et al. 2010). People also tend to be agreeable, or acquiesce, in interview situations (Schuman and Presser 1981).

Interviewer effects notwithstanding, over the past two decades, we have seen a steady decline in response rates for telephone interviews (de Leeuw 2008; de Leeuw and de Heer 2002). Moreover, with an increasing number of gated communities and a decreasing willingness to respond to in-person interviewers—in combination with the excessive costs—there are problems for researchers employing in-person methodologies (Groves and Couper 1998). With the telephone, the heavy use of answering machines as well as caller identification has presented formidable barriers to achieving contact with households and individuals (Groves and Lyberg 1988).

However, today an even more profound challenge has been the decrease in coverage for the traditional random digit dialing (RDD) samples, which only include landline telephone numbers (Blumberg et al. 2011). The decline in landline connections, particularly for young people, households with children, and people with changing social relationships such as separation and divorce, make coverage a greater problem (Keeter et al. 2007). Although mobile phone numbers are often added to landline sample frames to increase coverage, this dual frame approach introduces new challenges. For instance, dual sample frames are more prone to the sources of error in comparison to RDD-only frames. Moreover, in the United States, there are instances where incoming calls are counted against the respondent's prepaid minutes or data plans (Brick et al. 2011a). In addition, mobile phones produce geographic location challenges because individuals commonly keep the same mobile phone number after they relocate, rendering the area code a poor indicator of residential area.

The Internet

With the decline in interview response rates, there was much optimism that the Internet would become the survey replacement for the telephone. For many reasons, that has not happened. For instance, in the United States,

household Internet coverage is only about 75% (National Telecommunications and Information Administration [NTIA] 2011). Moreover, in those households with connections, some individuals lack the skills to use it, are uncomfortable with it, or use it infrequently (Stern et al. 2009).

Achieving a high response rate by e-mail contact only is also problematic. Specialized populations notwithstanding (e.g., attendees at conferences or recent doctorate graduates), very few e-mail-only surveys have been found effective at achieving high response rates. In addition, there is no acceptable general population sample frame for e-mail addresses, and professional norms prevent sending survey requests to individuals with whom there is no prior relationship (e.g., client, student, and association member). One can sometimes use mail (or even telephone) contact to ask people to respond over the web, but doing so adds to the cost and also has to be effective in convincing people to go from that mode of contact to a computer where they enter a URL and password (Anderson and Tancreto 2011).

An even bigger challenge is now emerging. There is evidence that significant numbers of people find they do not need a desktop or even a laptop computer (Miller et al. 2012). Instead, a smartphone (or other mobile device) is considered adequate for all of their electronic communication needs. For instance, Pew Internet and American Life Research Center (Miller et al. 2012:7) recently reported that nearly half of U.S. adults own a smart phone and 17% of them do the majority of their web searching on this device instead of on a computer. Asking questions on a smartphone requires utilizing formats that require very little space (Callegaro 2010). Thus, some traditional question formats (e.g., matrices) may be unusable when a significant portion of one's sample relies only on such devices (Buskirk and Andrus 2012).

Mail

Mail surveys have continued to do reasonably well both as a singular mode or when used in conjunction with another form of data collection despite the challenges discussed previously (Dillman et al. 2009). However, researchers have expressed several concerns.

First, it is now considered a higher cost mode of self-administered data collection in comparison to the web (Dillman et al. 2009) due to the expenditures associated with postage, processing, and printing (Groves et al. 2009).

A second reason for the reluctance to use mail surveys is that it is seen as less flexible given it is difficult to use branching or probe questions as effectively as in interviews or on the web, thus limiting the complexity and, in

some cases, depth of a survey instrument (Biemer et al. 2004; Bourque and Fielder 2003; Lyberg and Kasprzyk 1991). Furthermore, it takes longer to obtain responses, and evidence exists that responses are less complete compared to other modes (Dillman et al. 2009). In addition, item nonresponse is often slightly higher even in well-designed mail surveys when compared to web and telephone (Dillman 2012).

Finally, despite the availability of the U.S. Postal Service's Delivery Sequence File, which now makes attainable nearly all addresses of residential households where mail is delivered, there are still coverage and nonresponse issues. For example, similar to landline telephones, mail surveys have been shown not to obtain adequate response rates for: (1) younger people who do not use the mail system readily; (2) individuals in transitions such as when relocating; (3) those living at "drop points" (e.g., incomplete addresses for multiple family dwellings); (4) people living in new construction; and (5) illiterate and English as a Second Language (ESL) populations (Fowler 2009; Messer and Dillman 2011). Additionally, postal surveys suffer from being seen as junk mail when not adequately designed (Dillman et al. 2009).

In Search of Possible Solutions

Thus, there are many issues facing survey methodology today. This has led to a lot of new empirical work and, in some cases, nearly as many new questions. The research has offered us numerous solutions and possibilities, only a sample of which can be addressed in this overview.

Different Modes for Different Survey Situations

Unlike in the past, today it is clear that no single data collection mode is adequate for a majority of survey situations. That is, different populations, sponsorships, client needs, and circumstances require different modes and implementation procedures to reduce the major sources of survey error and keep survey costs in check.

Interview modes: Despite low response rates and coverage problems, we are going to continue to do single-mode interview surveys in some situations. Researchers will still use telephone-only surveys with mobile phone augmentation of landline lists, particularly for political polling due to the need for repeated timely results. In addition, they likely will still be used in surveys of the general population where the likelihood that a respondent will be eligible to respond is low, such as in the case of National Immunization Survey (NIS),

where the eligible respondent must have a child between the ages of 19 and 35 months and the eligibility rate is less than 3%. With millions of households screened in any given year, using mail in this case would be time inefficient. Thus, in some instances, there is simply no suitable alternative yet that gives the needed speed of reporting, cost efficiencies, and coverage (AAPOR Cell Phone Task Force 2010).

In election studies, we typically have exit poll data that allow statistical adjustment. However, in many cases, statistics produced through telephone-only surveys simply will not be convincing because of our lack of knowledge regarding the correlates for most data, which effectively prohibits post-stratification weighting. As a result, we will also continue to use in-person interviews as is done for the Current Population Survey and American Community Survey that collect critical societal information and thus cannot tolerate the coverage and nonresponse errors found with other modes.

Self-administered modes: In addition to the current challenges associated with interviewing discussed previously, there are two more significant reasons why self-administered modes are growing in popularity for general population surveys with mail perhaps being the predominant self-administered mode used for the near future. First, there is the availability of adequate mail sample frames (e.g., U.S. Postal Service's Delivery Sequence File and household registers in Europe). Second, there is still uneven diffusion of high-speed Internet access combined with low Internet proficiency among certain segments of the population and reluctance among some individuals to provide personal information online (Stern et al. 2009; Stern et al. 2012). What is more, household survey tests of mail-only survey methods show that it may obtain response rates of 40–70% in nonfederal government surveys (Messer and Dillman 2011; Smyth et al. 2010). Studies have also shown that mail can be effective for screening a household's characteristics and following up with a detailed set of questions, which is comparable with the two-stage telephone request (Brick et al. 2011b; Mathiowetz et al. 2010).

Are Shared Online Panels a Viable Alternative to One-time Samples?

One of the proposed solutions to the current challenges associated with individual mode surveys is to recruit panels of respondents who agree to complete a series of surveys over the Internet. The idea finds its basis in the work of Saris and de Pijper (1986) who, in 1984, began development of the first telepanel, a representative sample of Dutch citizens using computer-assisted surveying (Saris 1998). With the rise of the Internet over the past

two and a half decades, two sampling models have dominated the developments in this area.

In the nonprobability model, individuals are asked to join the study, most often when they are visiting a web site. The sponsoring organizations tend to offer per survey incentives to increase the attractiveness of panel membership. The approach has two issues with which it must contend. First, the samples are not representative with an overrepresentation of, for example, Internet users, students, and homemakers seeking to earn money. Second, when payments are the driving force for staying in the panel, the motivation for providing optimal answers must be questioned.

The second model relies on a probabilistic approach using RDD or ABS samples. Several research firms (e.g., CentERdata, InterSurvey, Gallup, and Knowledge Networks) have been using probabilistic methods in their Internet panels. Many of the current probabilistic online panel surveys follow the model of the CentERdata panels from the Netherlands, using a combination of traditional probability sampling and stratified refreshment samples to ensure representativeness (see Scherpenzeel and Das 2010 for a full description). For instance, Knowledge Network recruits their members via an ABS sampling approach and then provides Internet access and a laptop computer with the Windows operating system for the sampled individuals who do not have a computer or Internet access in their homes. Gallup recruits through RDD and then assigns respondents to web or mail based on user characteristics. Although these approaches potentially aim to solve the issue of web coverage, they are costly and dependent on web proficiency and preference. In addition, as far as we know, there is no experimental evidence that providing Internet access leads to more representative and better data among probability-based panels.

Both models present unique challenges. For instance, both probability- and nonprobability-based panels are subject to conditioning effects; however, we need further investigation whether this is more the case with Internet panels. Although research has shown that the accuracy of the probability-based online panels exceeds that of nonprobability online panels, they still are not able to obtain the level of precision of traditional methods such as RDD surveys (Chang and Krosnick 2009; Yeager et al. 2011).

Assessing the magnitude of the differences between traditional modes of survey data collection, including face-to-face and RDD surveys and online panels, requires consideration of two interdependent factors—nonrandom selection and differences between Internet users and the general population. As Chang and Krosnick (2009) and others have noted, when considering an opt-in and even some probability-based online panels as a survey data

source, it must be understood that not every member of the population has an equal chance for inclusion in the study. Recent attempts have been made to use various post-data adjustments, such as weighting, and sampling strategies to offset this nonrandom propensity for inclusion as well as examining various ways to assess the differences between online panels and other modes (e.g., absolute average error, largest absolute error, number of significant differences, etc.).[1] A confounding factor tends to be that Internet users still differ on key sociodemographic variables that lead to differences in their substantive responses; this is true on both behavioral and attitudinal measures (Duffy et al. 2005). For example, proficient Internet users tend to be more highly educated and wealthier, personal characteristics that covary with measures of political affiliation, views toward community change, and civic/political engagement (Stern and Dillman 2006). Although research has shown that some post-stratification strategies—including combined demographic and propensity weighting and alternative selective sampling approaches—can potentially offset some of the differences, this is not universally the case (Duffy et al. 2005; Terhanian and Bremer 2012).

Their representativeness notwithstanding, nonprobability surveys are inexpensive, and fast, which makes them appealing to marketing firms and researchers with a specific, targeted sample, but they are not based on well-established statistical or methodological theory (Dennis et al. 2011). We can envision strong opposition among many governmental and nongovernmental organizations to include their survey in someone else's panel, thus making this approach only feasible for select survey situations. Still, we are seeing an increasing number of these panels used in the marketing and nonprofit survey sector to supplement and augment existing modes when responses have not reached expectations; thus, more research in this area is certainly forthcoming (Terhanian and Bremer 2012).

When One Survey Mode Is Not Enough

Although mixing modes is not new (see de Leeuw 2005; Lyberg and Kasprzyk 1991; Payne 1964), it is only in the last decade that the strategy has become well established. This point is evidenced by the increasing number of empirically based studies on the effects of mixing modes as well as the more pervasive use of two or three modes by government and large survey organizations (Dillman et al. 2009). Recent research has helped bring to light effective ways and reasons for survey researchers to implement self-administered mixed-mode surveys.

One question that has arisen is whether different modes should be offered concurrently or sequentially. Studies have found that offering a concurrent choice in mode does little to improve response (Dillman et al. 2009; Gentry and Good 2008). In fact, Fulton and Medway (2012) found in their meta-analysis, which included 19 experimental comparisons, that offering a concurrent web option in mail surveys significantly reduced response rates. Similarly, Millar and Dillman (2011) found among college students that this concurrent approach did not improve response; but it did not significantly diminish response rates either. However, a sequential approach of web then mail did improved response rates.

Nonetheless, mixed-mode surveys can improve the representativeness of a sample. For instance, Messer and Dillman (2011) show that web-only approaches exclude important and often underrepresented segments of the population. However, they also found that web surveys supplemented by a subsequent mail questionnaire produced results similar to those of mail-only surveys, but at an increased monetary cost to the researcher.

There are some additional concerns researchers must consider in using a mixed-mode approach. For instance, mixing modes can lead to differences in substantive responses due to one mode favoring a certain sociodemographic over another mode (e.g., RDD attracts older homeowners whereas web-only surveys are more apt to obtain responses from younger people). Another source of differences stems from mode of presentation, with visual self-administered surveys often obtaining different answers than aurally administered questionnaires (de Leeuw and Hox 2011; Dillman and Christian 2005).

Additionally, with a web component, surveyors must attempt to understand the way different segments of society use the web. Whereas 96% of the U.S. population has access to and can use a landline telephone (NTIA 2011), web proficiency is much more variable. When using a web survey component, one must design for ease and the ability to replicate the design in another mode. We simply do not know what effects new possible question structures will have on measurement error in different segments of the population based on web proficiency. Finally, today we must consider how the visual design in a web component translates to mobile devices and the consequences for measurement error, response burden, and the interface of smartphone and web (Buskirk and Andrus 2012).

Accepting a Culture of Self-administration

The way people communicate is changing and surveyors need to adapt to that. As we have discussed previously, a substantial amount of research has

documented the increasing number of households without a landline telephone continues to rise, rendering traditional RDD frames incomplete (e.g., Blumberg and Luke 2012). In addition to this technological change, there has been a cultural shift away from synchronous communication and reliance on human intermediaries. This move is most often discussed in sociologically based theories such as "networked individualism" or "glocalization," where distance and localized personal communication is increasingly less needed for meaningful matters (e.g., Hampton and Wellman 2003; Stern and Dillman 2006; Wellman 2001). We now use automated systems to do much of our banking and bill paying as well as to get answers from local or governmental agencies, make reservations or make appointments. Perhaps this explains why we are less tolerant of interviews—they are increasingly nonnormative. There is some empirical evidence of this form in survey research. Smyth et al. (2010) included a question in a general population mixed-mode survey asking about the mode in which respondents preferred to complete a survey (telephone, web, or mail). Their results show a full 98.8% of respondents chose a self-administered mode.

One question to come out of this social change is whether interactive voice recognition (IVR) has a future (see Steiger and Conroy 2008). With IVR, people are given an automated aural stimulus and respond either aurally or using a keypad on their phones. Although the systems can be cumbersome, an entire generation does not recall when a live operator was the first level of connection. As a result, society is perhaps primed for this mode of delivery and response. In addition, in the developing world, many more people own mobile phones than have access to the Internet or a landline connection (Lerer et al. 2010). Technologically, the transition from aural to visual may be less pronounced than ever, thanks to smartphones. People across many generations are now avidly text messaging, meaning keypad entry is less problematic than it once was. Therefore, as smartphones increasingly become the normative tool of communication, IVR may be well situated. Recent research has shown that keypad entry as part of an IVR approach is possible when there is little mixing between the need to vocalize responses and manually enter them (Kuun 2009; Lerer et al. 2010). Although the future of IVR remains unclear, there are compelling reasons to continue to explore its utility.

Understanding How Visual Layout Affects Answers

Concomitant with the trend toward self-administered surveys has been the need to understand the way respondents are affected by the visual design of questions and the questionnaire. Twenty years ago, survey methodologists focused more

on the wording of questions than overall questionnaire design and how people responded to it. During the 1990s, cognitive interviewing became the primary means for evaluating both questions and questionnaires (Lessler and Forsyth 1996; Tourangeau et al. 2000). Survey methodologist began to uncover how small manipulations in the visual design of questions influenced respondents' answers (Jenkins and Dillman 1997). Over the past decade, research has shown that the visual design of a survey and its various components are as influential to responses as any documented interviewer or wording effect.

The body of experimental work in this area has shown that visual layout may dramatically affect the reporting of numbers and selection of response categories; font size, spacing, location of information, and symbols affect answers. For example, Christian and Dillman (2004), Smyth et al. (2006), and Stern et al. (2007) have all shown how arrows and answer boxes operate independently and in concert to communicate information to respondents on the navigational path required for completing self-administered questionnaires. In related work, Tourangeau et al. (2004, 2006) have shown that there are interpretive heuristics based on spatial or visual features that influence respondents.

Putting this research in perspective brings to light two important issues for the future. First, in a time when mixed-mode surveys are seen as a necessity as opposed to a novelty, we must be cognizant of differences in the visual design across different modes and platforms. Survey researchers simply cannot design a web and paper version of the survey; instead, there must be an attempt at common design. Second, web survey designers must now evaluate how their questionnaires appear on smartphones and tablets. Not doing so is no longer option.

Discussion

In Greek mythology, the two-headed Janus figure is used as a symbol of change and transition as well as the need to simultaneously see the past and the future. This symbolism is appropriate as we attempt to understand the current state of survey methodology. In dealing with the challenges of surveying today, we cannot solely rely on what we know has worked in the past while ignoring the vast technological and societal changes that are part of our world. Nor can we solely rely on what is coming toward us without understanding the principles of effective surveying we have developed through decades of research. An integrative approach where we look to the future with an open mind while remaining ever conscious of the lessons we have learned from our past will determine our success in this ever-changing environment. Three themes emerge from this overview, which highlight areas of research and development to consider as we move forward.

First, whether we are discussing the mixing of modes, recruiting of panels, or designing a survey in a way that the visual aspects work across multiple devices, we must design for the population of interest. Schaeffer and Dykema (2011:930) suggest that increasingly "survey design may become more heterogeneous because of varying research needs and development of diverse technologies." Mobile devices are increasingly replacing desktop and even laptop computers as the primary way people access the web. However, this is not happening at an even demographic pace: Similar to adoption of the Internet, younger users are the early adopters. As a result, a survey catered to, for example, recent college graduates, should probably be designed with a great deal of attention given to the effects of technology on the visual design. On the other hand, a survey of school administrators, which could very well be on the same topic, should be more concerned with the appropriateness of modes. This means that larger, general population studies face unprecedented challenges. Within these studies, the best strategy for collecting data from large and important segments of the population might be the worst way to survey others. However, within one study, multiple approaches can be applied. For instance, the implications of and best practices for using web and mail or IVR in some combination are still being uncovered and we need more experimental work.

Second, with the rise in self-administered surveys and declining response rates for interview modes (Couper 2011), the increasing interest in Internet panels (Chang and Krosnick 2009; Yeager et al. 2011), the intrigue surrounding the potential of social network sites (Bhutta 2012; Matulewicz et al. 2012), and search engine sampling (Nunan and Knox 2011), nonprobability methods will continue to gain attention. Their practicality in a time of change makes them attractive and they are being used with considerable frequency. We find examples of the growing interest in nonprobability methods in the keynote talks by Doug Rivers and George Teherian at the 2012 AAPOR Conference. Both presentations discussed nonprobability sampling methods raising concerns about the guiding theoretical premise for using these methods. There is much work to be done in this area. The limitations and potentialities of nonprobability methods should be studied further.

The third theme to emerge from this overview is that our traditional view of discrete approaches, in particular modes, may now be antiquated for many studies. As society and technology have changed, the boundaries between traditional modes have become less clear. For example, although a telephone and mail surveys may still be very distinct and definable as aural and visual, respectively, there are many integrative possibilities for surveys geared toward completion on web, smartphone, or tablet platforms

(e.g., videos or avatars). In a relatively short amount of time, the term "web survey" has morphed from a mode to a "multidimensional" concept (Couper 2011:892). Thus, a web instrument could mean a number of things. As we noted previously, research has also shown that completion of IVR surveys with a keypad serves as an intermediary between self-administered and interview surveys, between visual design and aural comprehension. The literature reviewed here certainly suggests that understanding other places where these "bridges" exist will be important to improve response rates while controlling the various types of errors and survey costs.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Note

1. See Duffy et al. (2005), Yeager et al. (2011), and Terhanian and Bremer (2012) for full discussions and examples.

References

AAPOR Cell Phone Task Force. 2010. New considerations for survey researchers when planning and conducting RDD telephone surveys in the U.S. with respondents reached via cell phone numbers. *AAPOR Cell Phone Task Force Report 1*. http://www.aapor.org/AM/Template.cfm?Section=Cell_Phone_Task_Force_Report&Template=/CM/ContentDisplay.cfm&ContentID=3189 (accessed May 15, 2012).

Anderson, A., and J. Tancreto. 2011. Using the web for data collection at the United States Census Bureau. http://isi2011.congressplanner.eu/pdfs/450221.pdf (accessed February 13, 2012).

Belli, R., M. Traugott, M. Young, and K. McGonagle. 1999. Reducing vote overreporting in surveys: Social desirability, memory failure, and source monitoring. *Public Opinion Quarterly* 63:90–108.

Bhutta, C. 2012. Not by the book: Facebook as a sampling frame. *Sociological Methods & Research* 41:57–88.

Biemer, P., R. M. Groves, L. E. Lyberg, N. A. Mathiowetz, and S. Sudman. 2004. *Measurement errors in surveys*. New York: John Wiley.

Blumberg, S., and J. Luke. 2012. Wireless substitution: Early release of estimates from the National Health Interview Survey, July–December 2011. *National Center for Health Statistics.* http://www.cdc.gov/nchs/data/nhis/earlyrelease/wireless201005.htm (accessed February 12, 2012).

Blumberg, S., J. Luke, N. Ganesh, M. Davern, M. H. Boudreaux, and K. Soderberg. 2011. Wireless substitution: State-level estimates from the National Health Interview Survey, January 2007–June 2010. *National Health Statistics Report 29*. http://www.ncbi.nlm.nih.gov/pubmed/21568134 (accessed February 12, 2012).

Bourque, L. B., and E. P. Fielder. 2003. *How to conduct self-administered and mail surveys*. 2nd ed. Newbury Park, CA: Sage.

Brick, M., D. Williams, and J. Montaquila. 2011. Addressed-based sampling for subpopulations. *Public Opinion Quarterly* 75:409–28.

Buskirk, T. D., and C. Andrus. 2012. Smart surveys for smartphone: Exploring various approaches for conducting online mobile surveys via smartphones. *Survey Practice*. http://surveypractice.wordpress.com/2012/02/21/smart-surveys-for-smart-phones/ (accessed April 20, 2012).

Callegaro, M.2010. Do you know which device your respondent has used to take your online survey? *Survey Practice*. http://surveypractice.wordpress.com/2010/12/08/device-respondent-has-used/ (accessed November 8, 2011).

Chang, L., and J. Krosnick. 2009. National surveys via RDD telephone interviewing versus the Internet: Comparing sample representativeness and response quality. *Public Opinion Quarterly* 734:641–78.

Christian, L. M., and D. A. Dillman. 2004. The influence of symbolic and graphical language manipulations on answers to paper self-administered questionnaires. *Public Opinion Quarterly* 68:57–80.

Couper, M. 2011. The future of modes of data collection. *Public Opinion Quarterly* 75:889–908.

Davis, C. G., J. Thake, and N. Vilhena. 2010. Social desirability biases in self-reported alcohol consumption and harms. *Addictive Behaviors* 354:302–11.

de Leeuw, E. D. 2005. To mix or not to mix data collection modes in surveys. *Journal of Official Statistics* 21:233–55.

de Leeuw, E. D. 2008. Choosing the method of data collection. In *International handbook of survey methodology*, eds. E. D. de Leeuw, J. J. Hox, and D. A. Dillman, 113–35. London: Taylor & Francis.

de Leeuw, E. D., and W. de Heer. 2002. Trends in household survey nonresponse: A longitudinal and international comparison. In *Survey nonresponse*, eds. R. M. Groves, D. A. Dillman, J. L. Eltinge, and R. J. A. Little, 41–54. New York: Wiley.

de Leeuw, E. D., and J. Hox. 2011. Internet surveys as part of a mixed mode design. In *Social and behavioral research and the Internet: Advances in applied methods*

and new research strategies, eds. M. Das, P. Ester, and L. Kaczmirek, 45–76. New York: Routledge, Taylor and Francis Group.

Dennis, J., Y. Kruse, and T. Thompson. 2011. Examination of panel conditioning effects in a web-based 2008 election study. Paper presented at the 66th Annual Conference of American Association for Public Opinion Research, Phoenix, May 12–15.

Dillman, D. A. 2002. Navigating the rapids of change: Some observations on survey methodology in the early twenty-first century. *Public Opinion Quarterly* 66:473–94.

Dillman, D. A. 2012. Introduction to Special Issue of Survey Practice on item nonresponse. *Survey Practice* 5:1–3.

Dillman, D. A., and L. Christian. 2005. Survey mode as a source of instability in responses across surveys. *Field Methods* 17:30–51.

Dillman, D. A., J. Smyth, and L. Christian. 2009. *Internet, mail, and mixed-mode surveys: The tailored design method*. Hoboken, NJ: John Wiley.

Duffy, B., K. Smith, G. Terhanian, and J. Bremer. 2005. Comparing data from online and face-to-face surveys. *International Journal of Market Research* 47:615–39.

European Association of Methodology Series. New York: Lawrence Erlbaum Associates.

Fowler, F. J. 2009. *Survey research methods*. Thousand Oaks, CA: Sage.

Fulton, J., and R. Medway. 2012. When more gets you less: A meta-analysis of the effect of concurrent web options on mail survey response rates. Presented at the 67th Annual Conference of American Association for Public Opinion Research, Orlando, May 17–20.

Gentry, R., and C. Good. 2008. Offering respondents a choice of survey mode: Use patterns of an Internet response option in a mail survey. Presentation at the Annual Conference of the American Association of Public Opinion Research, New Orleans, May 15–18.

Groves, R. M., and M. P. Couper. 1998. *Nonresponse in household interview surveys*. New York: Wiley.

Groves, R. M., F. J. Fowler, M. P. Couper, J. M. Lepkowski, E. Singer, and R. Tourangeau. 2009. *Survey methodology*. 2nd edition. Hoboken, NJ: John Wiley.

Groves, R. M., and L. Lyberg. 1988. An overview of nonresponse issues in telephone surveys. In *Telephone survey methodology*, eds. R. Groves, P. Biemer, L. Lyberg, J. Massey, W. Nicholls, and J. Waksberg. 191–212. New York: John Wiley & Sons.

Hampton, K., and B. Wellman. 2003. Neighboring in Netville: How the Internet helps connect people in a wired suburb. *City and Community* 2:277–311.

Jenkins, C., and D. A. Dillman. 1997. Towards a theory of self-administered questionnaire design. In *Survey nonresponse*, eds. R. Groves, D. A. Dillman, J. Eltinge, and R. Little, 165–96. New York: Wiley.

Keeter, S., C. Kennedy, A. Clark, T. Tompson, and M. Mokrzycki. 2007. What's missing from national landline RDD surveys?: The impact of the growing cell-only population. *Public Opinion Quarterly* 715:772–92.

Kish, L. 1965. *Survey sampling*. New York: Wiley.

Kuun, C. 2009. OpenPhone project piloted in Botswana. http://www.csir.co.za/enews/2008_july/ic_05.html (accessed January 14, 2012).

Lerer, A., M. Ward, and S. Amarsinghe. 2010. Evaluation of IVR data collection UIs for untrained rural users. Proceedings of the First ACM symposium on computing for development, ACM DEV'10, December 17–18. London, UK: ACM Press. http://dspace.mit.edu/handle/1721.1/73000 (accessed July 17, 2012).

Lessler, J., and B. Forsyth, 1996. A coding system for appraising questionnaires. In *Answering questions: Methodology for determining cognitive and communicative processes in survey research*, eds. N. Schwartz and S. Sudman, 259–92. San Francisco, CA: Jossey-Bass.

Lyberg, L. E., and D. Kasprzyk. 1991. Data collection methods and measurement error: An Overview. In *Measurement errors in surveys*, eds. P. P. Biemer, R. M. Groves, L. E. Lyberg, N. A. Mathiowetz, and S. Sudman, 237–57. New York: Wiley.

Mathiowetz, N., M. Brick, L. Stokes, R. Andrews, and S. Muzzy. 2010. A pilot test of a dual-frame mail survey as an alternative to an RDD survey. Paper presented at the Joint Statistical Meetings, Vancouver, BC, July 28–August 2.

Matulewicz, H., D. Friend, S. Boraas, and A. Ciemnecki. 2012. Respondent permission to contact or locate on Facebook: Preliminary findings from the National Longitudinal Transition Study 2012. Presented at the 67th Annual Conference of American Association for Public Opinion Research, Orlando, May 17–20.

Messer, B., and D. A. Dillman. 2011. Surveying the general public over the Internet using address-based sampling and mail contact procedures. *Public Opinion Quarterly* 75:429–57.

Millar, M., and D. A. Dillman. 2011. Improving response to web and mixed-mode surveys. *Public Opinion Quarterly* 75:249–69.

Miller, C., K. Purcell, and T. Rosenstiel. 2012. 72% of Americans follow local news closely. *Pew Research Center*. http://pew.org/Reports/2012/Local-news-enthusiasts.aspx (accessed July 20, 2012).

NTIA (National Telecommunications and Information Administration). 2011. Exploring the digital nation: Computer and Internet use at home. Report prepared the U.S. Department of Commerce. http://www.ntia.doc.gov/files/ntia/publications/exploring_the_digital_nation_computer_and_internet_use_at_home_11092011.pdf (accessed February 8, 2012).

Nunan, D., and S. Knox. 2011. Can search engine advertising help access rare samples? *International Journal of Market Research* 53:523–40.

Payne, S. 1964. Combination of survey methods. *Journal of Marketing Research* 12: 61–62.

Saris, W. E. 1998. Ten years of interviewing without interviewers: The telepanel. In *Computer assisted survey information collection*, eds. M. P. Couper, R. P. Baker, J. Bethlehem, C. Clark, J. Martin, W. L. Nicholls II , and J. M. O'Reilly, 409–31. New York: Wiley.

Saris, W. E., and W. M. de Pijper. 1986. Computer assisted interviewing using home computers. *European Research* 14:144–50.

Schaeffer, N., and J. Dykema. 2011. Questions for surveys current trends and future directions. *Public Opinion Quarterly* 75:909–61.

Scherpenzeel, C. A., and M. Das. 2010. True longitudinal and probability-based Internet panels: Evidence from the Netherlands. In *Social and behavioral research and the Internet: Advances in applied methods and new research strategies*, eds. M. Das, P. Ester, and L. Kaczmirek, 77–104. New York: Routledge, Taylor and Francis Group.

Schuman, H., and S. Presser. 1981. *Questions and answers in attitude surveys: Experiments on question form, wording, and context*. Orlando, FL: Academic Press.

Smyth, J., D. A. Dillman, L. Christian, and A. O'Neill. 2010. Using the Internet to survey small towns and communities: Limitations and possibilities in the early 21st century. *American Behavioral Scientist* 53:1423–48.

Smyth, J., D. A. Dillman, L. Christian, and M. Stern. 2006. Comparing check-all and forced-choice question formats in web surveys. *Public Opinion Quarterly* 70: 66–77.

Steiger, D., and B. Conroy. 2008. IVR: Interactive Voice Response. In *International handbook of survey methodology*, eds. E. D. de Leeuw and D. A. Dillman, 285–98. New York: Lawrence Erlbaum.

Stern, M. J., A. Adams, and S. Elsasser. 2009. Digital inequality and place: The effects of technological diffusion on Internet proficiency and usage across rural, suburban, and urban counties. *Sociological Inquiry* 79:391–417.

Stern, M. J., S. Cotten, and P. Drentea. 2012. The separate spheres of online health information searching: The influence of gender and parenting on behaviors and feelings about searching for health information on the web. *Journal of Family Issues* 33:1324–50.

Stern, M. J., and D. A. Dillman. 2006. Community participation, social ties, and use of the Internet. *City and Community* 5:409–24.

Stern, M. J., D. A. Dillman, and J. D. Smyth. 2007. Visual design, order effects and respondent characteristics in a self-administered survey. *Survey Research Methods* 1:121–38.

Terhanian, G., and J. Bremer. 2012. A smarter way of selecting respondents for surveys? *International Journal of Market Research* 26:751–80.

Tourangeau, R., M. Couper, and F. Conrad. 2004. Spacing, position, and order interpretive heuristics for visual features of survey questions. *Public Opinion Quarterly* 68:368–93.

Tourangeau, R., M. Couper, and F. Conrad. 2006. Color, labels, and interpretive heuristics for response scales. *Public Opinion Quarterly* 71:91–112.

Tourangeau, R., L. Rips, and K. Rasinski. 2000. *The psychology of survey response*. Cambridge, UK: Cambridge University Press.

Valliant, R., A. Dorfman, and R. Royall. 2000. *Finite population sampling and inference: A prediction approach*. New York: John Wiley.

Wellman, B. 2001. Computer networks as social networks. *Science* 293:2031–34.

Yeager, D. S., J. A. Krosnick, L. Chang, H. S. Javitz, M. S. Levendusky, A. Simpser, and R. Wang. 2011. Comparing the accuracy of RDD telephone surveys and Internet surveys conducted with probability and nonprobability samples. *Public Opinion Quarterly* 75:709–47.